Phishing Investigation – Case 01

Greenholt Phish | SOC L1 Investigation Report

Case ID	PHISH-001
Incident Type	Phishing Email
Severity	HIGH
Initial Detection	Suspicious email reported by a user
Primary Attack Vector	Spearphishing Attachment
MITRE ATT&CK	T1566.001 – Spearphishing Attachment
Status	Investigated – Containment Recommended

1. Alert Summary

A suspicious phishing email was reported by a user. The email claimed that funds had been transferred through SWIFT and included a payment reference number. A suspicious attachment was included with the message.

2. Sender Analysis

Sender	info@mutawamarine.com
Sender Domain	mutawamarine.com
Display Name	Mr. James Jackson
Reply-To	Mr. James Jackson <info.mutawamarine@mail.com>

The Reply-To address uses a different domain from the sender address, which is an additional suspicious indicator.

3. Email Analysis

Subject: webmaster@redacted.org your: Transfer Reference Number:(09674321)

Attachment: SWT_#09674321____PDF__.CAB

Suspicious indicators:

• The email uses a financial/payment-transfer theme that may be intended to create urgency and credibility.

• The sender domain requires further investigation.

• The Reply-To address uses a different domain from the sender address.

• The attachment uses a .CAB extension while including PDF in its filename, which could be intended to make the file appear less suspicious.

• SPF authentication failed for the sender domain.

• The source IP was associated with hosting infrastructure.

• No URL was identified in the email body.

4. Header Analysis

SPF	FAIL

DKIM	No DKIM result found in the provided headers
DMARC	UNKNOWN
Source IP	192.119.71.157
IP Provider/ISP	HostPapa
ASN	AS54290

SPF evidence: `spf=fail smtp.mailfrom=mutawamarine.com`

SPF record identified: `v=spf1 include:spf.protection.outlook.com -all`

Received header: `Received: from hwsrv-737338.hostwindsdns.com ([192.119.71.157]:51810 helo=mutawamarine.com)`

The sending IP was not authorized by the domain's SPF policy based on the observed authentication result.

5. URL Analysis

No clickable URL was identified in the provided email content. The primary suspicious object in this case was the email attachment.

6. IOC Extraction

IOC Type	Indicator
IP	192.119.71.157
Domain	mutawamarine.com
URL	None identified
Attachment	SWT_#09674321____PDF__.CAB
SHA-256	2e91c533615a9bb8929ac4bb76707b2444597ce063d84a4b33525e25074fff3f

These indicators can be used to search for related activity across email systems, endpoint logs, DNS logs, proxy logs, and SIEM data.

7. Threat Intelligence

Reputation: Pending VirusTotal verification.

The attachment hash should be checked against VirusTotal or another trusted threat-intelligence source to determine whether the file has previously been identified as malicious.

The email already contains several suspicious indicators, including an SPF failure, suspicious sending infrastructure, a mismatched Reply-To address, and a .CAB attachment disguised with PDF in its filename.

```
SHA-256: 2e91c533615a9bb8929ac4bb76707b2444597ce063d84a4b33525e25074fff3f
```

8. MITRE ATT&CK; Mapping

Technique	T1566 – Phishing
Sub-technique	T1566.001 – Spearphishing Attachment
Rationale	The phishing email used an attachment as the primary mechanism for delivering potentially malicious content.

9. Severity Assessment

Severity: HIGH

The email was classified as high severity because it contains multiple indicators of a potential phishing attack. The message uses a financial-transfer theme, contains a suspicious attachment, has an SPF authentication failure, uses a different Reply-To domain, and originates from suspicious hosting infrastructure.

The attachment is particularly concerning because it uses a .CAB extension while including PDF in the filename, which could be intended to make the file appear less suspicious to the recipient.

If the attachment was opened, further endpoint investigation would be required to determine whether malware was executed or whether the user's system was compromised.

10. Recommended SOC Response

• Quarantine the phishing email.

• Prevent the attachment from being delivered to additional users.

• Block the malicious sender/domain if confirmed malicious.

• Search the mail environment for the same sender, subject, attachment name, and SHA-256 hash.

• Search endpoint and SIEM logs for the SHA-256 hash.

• Check whether any users opened or executed the attachment.

• Investigate any endpoint that interacted with the attachment.

• Check for suspicious processes, files, persistence mechanisms, and network connections.

• Reset credentials if there is evidence that credentials were exposed.

• Add confirmed IOCs to appropriate security controls.

• Escalate to L2/Incident Response if user interaction or compromise is confirmed.

11. Analyst Conclusion

The investigation identified multiple indicators consistent with a phishing attempt, including failed SPF authentication, a mismatched Reply-To address, suspicious sending infrastructure, and an attachment using a .CAB extension with PDF in its filename. The attachment hash and other IOCs should be checked against trusted threat-intelligence sources and searched across the organization's security telemetry. Based on the available evidence, the email should be treated as high-risk and contained while further endpoint investigation is performed if the attachment was opened.

Portfolio note: This report documents a simulated/lab-based SOC investigation and should be clearly labeled as such when presented to recruiters.